

# Overview of Quasi-Experimental Designs

Advanced Social Epidemiology PhD Course  
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**Epidemiology, Biostatistics  
and Occupational Health**

# Overview

Motivation

Quasi-Experimental Designs

Final Thoughts

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Quasi-Experimental Designs

Final Thoughts

# What's the problem?

- We are mainly (though not exclusively) interested in causal effects.
- We want to know:
  - Should we intervene to reduce exposure to  $X$ ?, or
  - Did the program work? If so, for whom? If not, why not?, or
  - If we implement the program elsewhere, should we expect the same result?
- These questions involve counterfactuals about what would happen if we intervened to do something.
- These are causal questions.

# Causation, Association, and Confounding

- Causal effect: Difference between those randomly **assigned** to treatment:

$$E(Y|SET[Treated]) - E(Y|SET[Untreated])$$

- Association: Difference between those who **happen to be** treated:

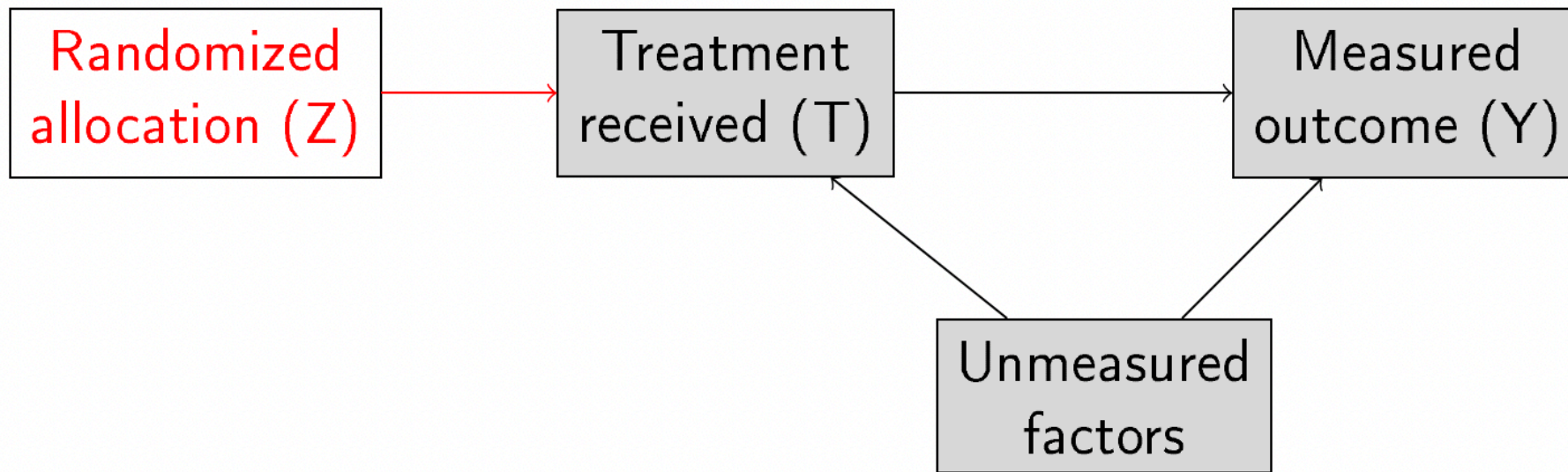
$$E(Y|Treated) - E(Y|Untreated)$$

- Confounding:

$$\begin{aligned} E(Y|SET[Treated]) - E(Y|SET[Untreated]) \\ \neq \\ E(Y|Treated) - E(Y|Untreated) \end{aligned}$$

## Strength of randomized treatment allocation

- Recall that randomization means that we can generally estimate the causal effect *of being randomized* without bias.
- Randomization guarantees exchangeability on measured and unmeasured factors.



## Randomize if you can

- Randomization leads to:
  - balance on measured factors.
  - balance on unmeasured factors.
- Unmeasured factors cannot bias the estimate of the exposure effect.
- Example from Home Injury Prevention Intervention cluster RCT
- What do you notice about Table 1?

Keall et al. (2015)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Treatment group<br>(n=950) | Control group<br>(n=898) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------|
| Female sex                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 541 (57%)                  | 501 (56%)                |
| Indigenous Māori                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 88 (9%)                    | 86 (10%)                 |
| Mean (SD) age (years)*                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 45 (28.0)                  | 43 (28.1)                |
| Age range (years)                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0–94                       | 0–92                     |
| 0–9                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 175 (18%)                  | 187 (21%)                |
| 10–19                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 89 (9%)                    | 82 (9%)                  |
| 20–29                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 34 (4%)                    | 37 (4%)                  |
| 30–39                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 116 (12%)                  | 112 (12%)                |
| 40–49                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 90 (9%)                    | 96 (11%)                 |
| 50–59                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 65 (7%)                    | 51 (6%)                  |
| 60–69                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 132 (14%)                  | 105 (12%)                |
| ≥70                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 249 (26%)                  | 228 (25%)                |
| Number of injuries at home, excluding falls, in past year (per person)†                                                                                                                                                                                                                                                                                                                                                                                                      | 122 (0.129)                | 103 (0.115)              |
| Number of fall injuries at home in past year (per person)‡                                                                                                                                                                                                                                                                                                                                                                                                                   | 87 (0.092)                 | 61 (0.068)               |
| Number of specific injuries in past year (per person)§                                                                                                                                                                                                                                                                                                                                                                                                                       | 23 (0.024)                 | 24 (0.027)               |
| Data are number of individual occupants (%), unless otherwise indicated.<br>*At Aug 3, 2010. †Injuries arising in the home during the 365-day period before the intervention date, obtained from matched insurance claim data. ‡Slips, trips, or fall injuries in the home during the 365-day period before the intervention date. §Injuries most specific to the package of home modifications, arising in the home during the 365-day period before the intervention date. |                            |                          |

**Table 1: Characteristics of individual occupants at baseline**

# Or maybe don't randomize?

## RCT limitations

- Non-compliance.
- Attrition.
- Spillovers.
- Blinding (esp. in clinical trials).

## Other trial challenges:

- Unethical (poverty, parental social class, job loss)
- Impossible (ethnic background, place of birth)
- Expensive (neighborhood environments)
- Long latency periods (many years before outcomes are observable).
- We need alternatives to RCTs.

## Key issue is credibility

- If we have a good design and assume that we have measured all of the confounders, then regression can give us exactly what we want: an estimate of the causal effect of exposure to  $T$ .
- Core issue: How credible is this assumption?



## Why we worry about observational studies

Recent evaluation of “Workplace Wellness” program in US state of Illinois

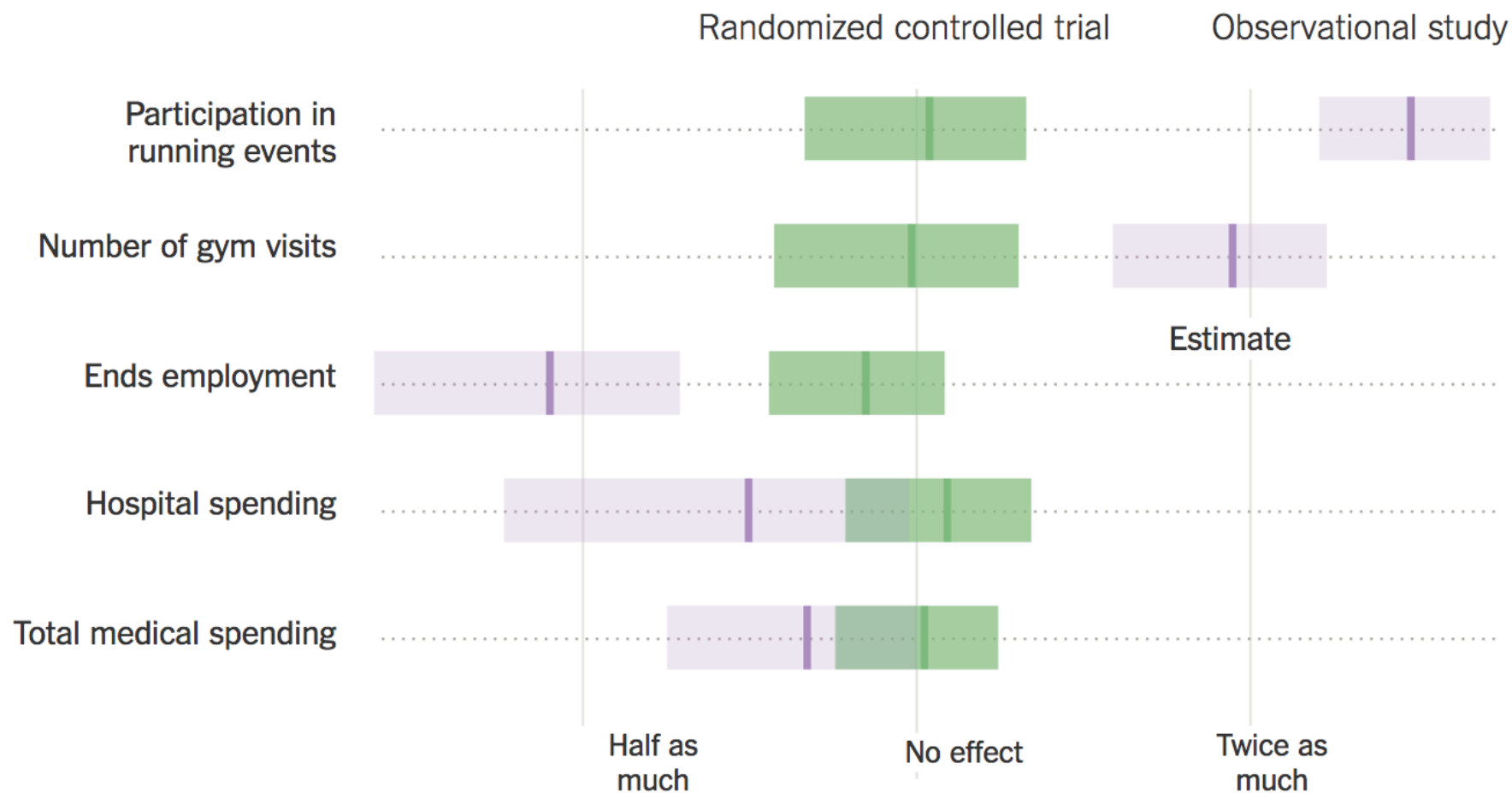
Treatment: biometric health screening; online health risk assessment, access to a wide variety of wellness activities (e.g., smoking cessation, stress management, and recreational classes).

Randomized evaluation:

- 3,300 individuals assigned treated group.
- 1,534 assigned to control (could not access the program).

Also analyzed as an observational study comparing “participants” vs. non-participants in treated group.

# How the Illinois Wellness Program Affected ...



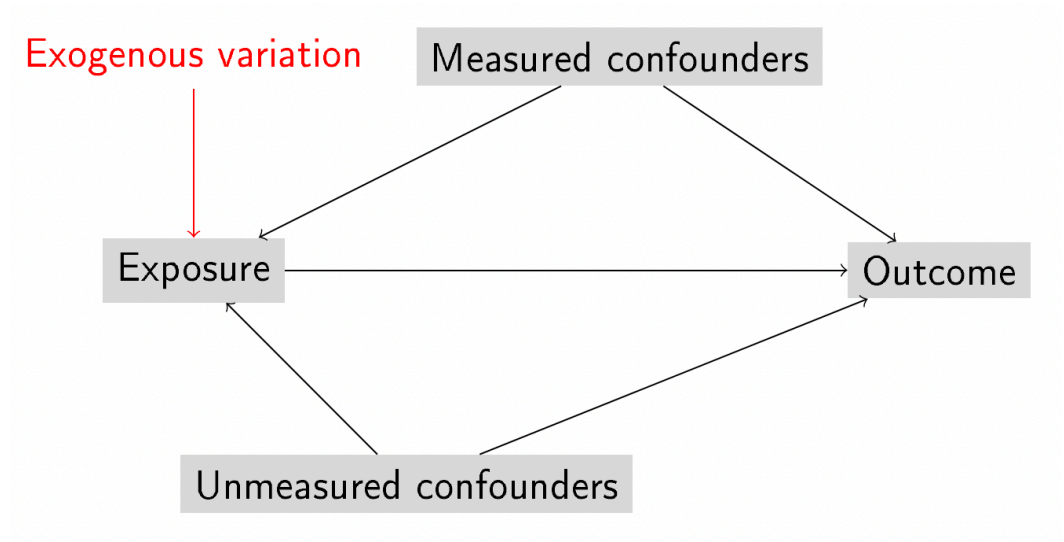
# How can quasi-experiments help?

Quasi-experiments aim to mimic RCTs.

- “Accidents of chance” that create:
  1. Comparable treated and control units
  2. Random or “as-if” random assignment to treatment.
- Control for (some) sources of bias that cannot be adequately controlled using regression adjustment.
- More credible designs also help us to understand the relevance of other factors that may be implicated in generating inequalities.

# Selection on “observables” and “unobservables”

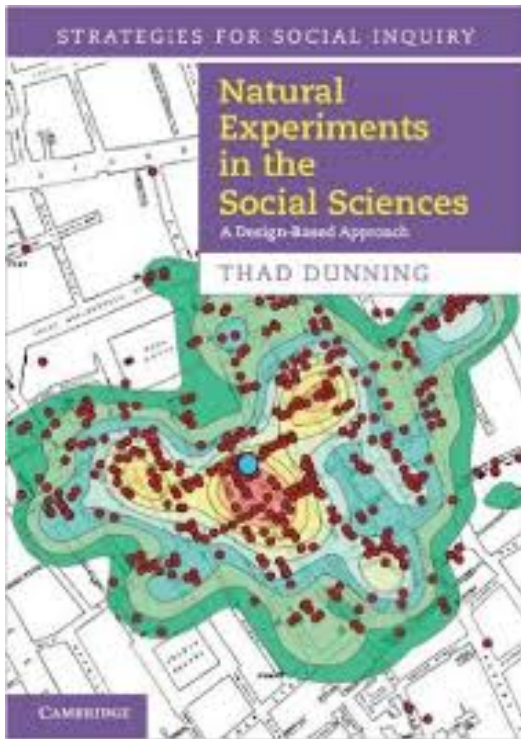
- Observables: Things you measured or can measure
- Unobservables: Things you can't measure (e.g., innate abilities, motivation)
- Exogenous variation: predicts exposure but (we assume) **not** associated with anything else [mimicking random assignment].



## Strategies based on observables and unobservables

- Most observational study designs control for *measured* factors using:
  - Stratification
  - Regression adjustment
  - Matching (propensity scores, etc.)
- Quasi-experimental strategies **aim** to control for some *unmeasured* factors using:
  - Interrupted time series (ITS)
  - Difference-in-differences (DD)
  - Synthetic controls (SC)
  - Instrumental variables (IV)
  - Regression discontinuity (RD)

# Some *potential* sources of natural experiments



- Law changes
- Eligibility for social programs (roll-outs)
- Lotteries
- Genes
- Weather shocks (rainfall, disasters)
- Arbitrary policy or clinical guidelines (thresholds)
- Business / factory closures
- Historical legacies (physical environment)
- Seasonality

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# Difference-in-Differences

# Difference-in-Differences: Basic Idea

In the simplest DD setting, outcomes are observed for units in two groups and in two time periods.

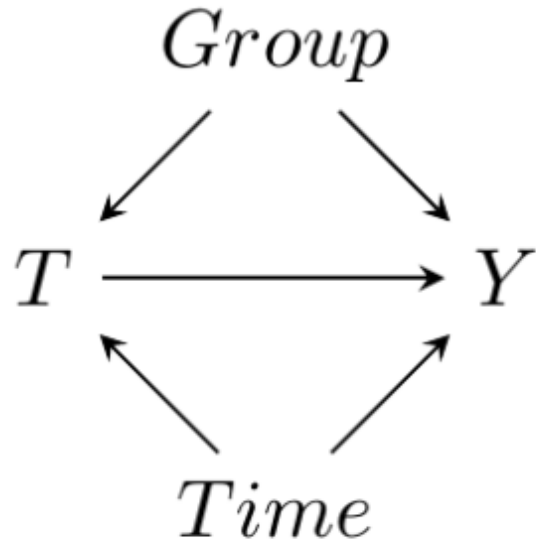
## Treated:

- Only units in one of the two groups are exposed to a treatment, in the second time period.

## Control:

- Never observed to be exposed to the treatment.

# Difference-in-Differences: Basic Idea

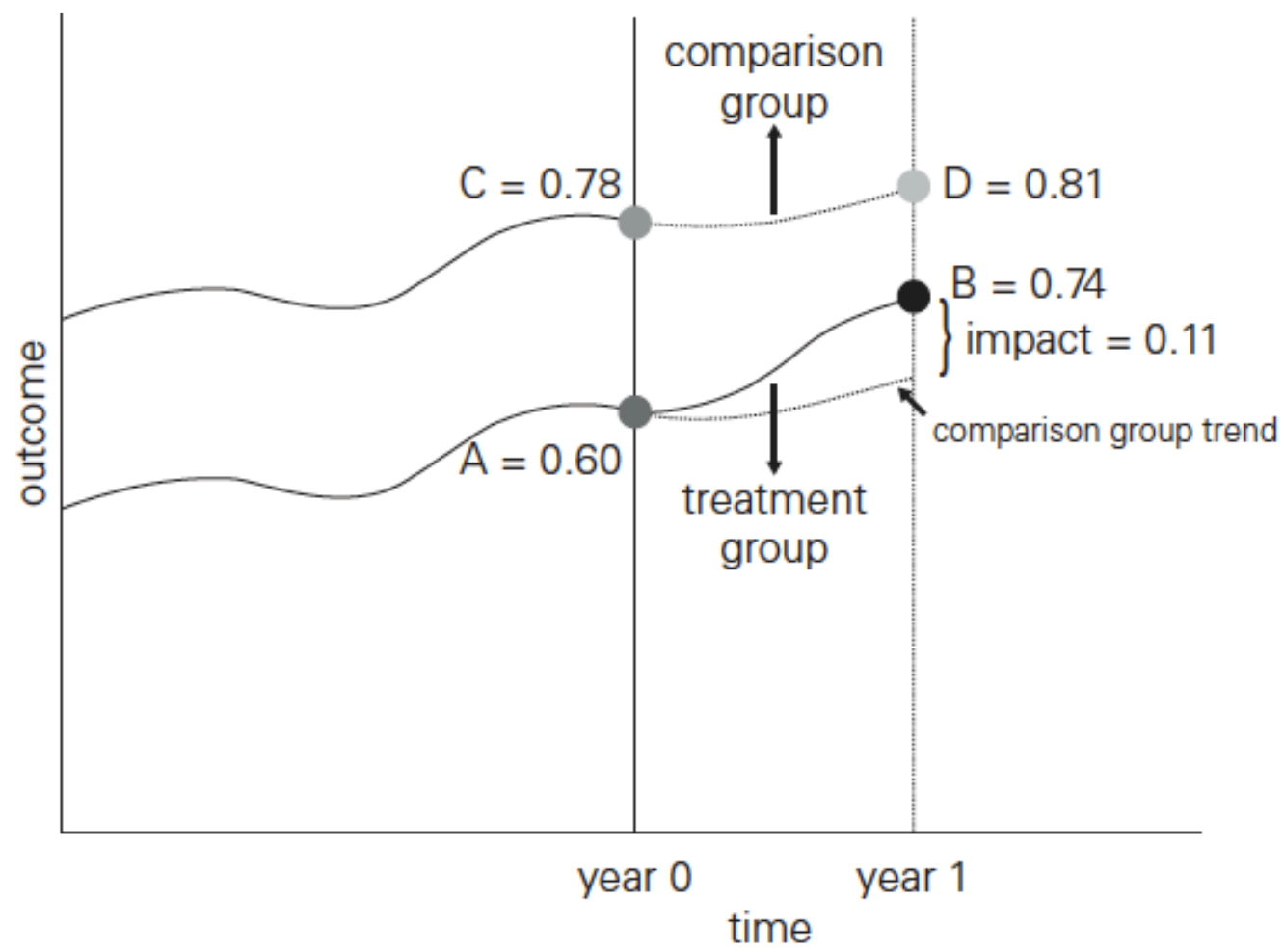


The average **change** over time (first difference) in the non-exposed (control) group is **subtracted** (second difference) from the gain over time in the exposed (treatment) group.

Double differencing removes biases in post-period that could result from *unmeasured*:

- fixed differences between those groups
- secular trends affecting both groups.

# Visual Intuition of DD



Gertler et al. (2011)

## Difference-in-Differences without Regression

DD is just differences in means! Let  $\mu_{it} = E(Y_{it})$

- $i = 0$  is control group,  $i = 1$  is treatment.
- $t = 0$  is pre-period,  $t = 1$  is post-period.
- One 'difference' estimate of causal effect is:  $\mu_{11} - \mu_{10}$  (pre-post in treated)
- Differences-in-Differences estimate of causal effect is:  $(\mu_{11} - \mu_{10}) - (\mu_{01} - \mu_{00})$

| Area    | Before | After | Difference (A - B) |
|---------|--------|-------|--------------------|
| Treated | 135    | 100   | -35                |
| Control | 80     | 60    | -20                |
| T - C   | 55     | 40    | -15                |

# A social epidemiology example

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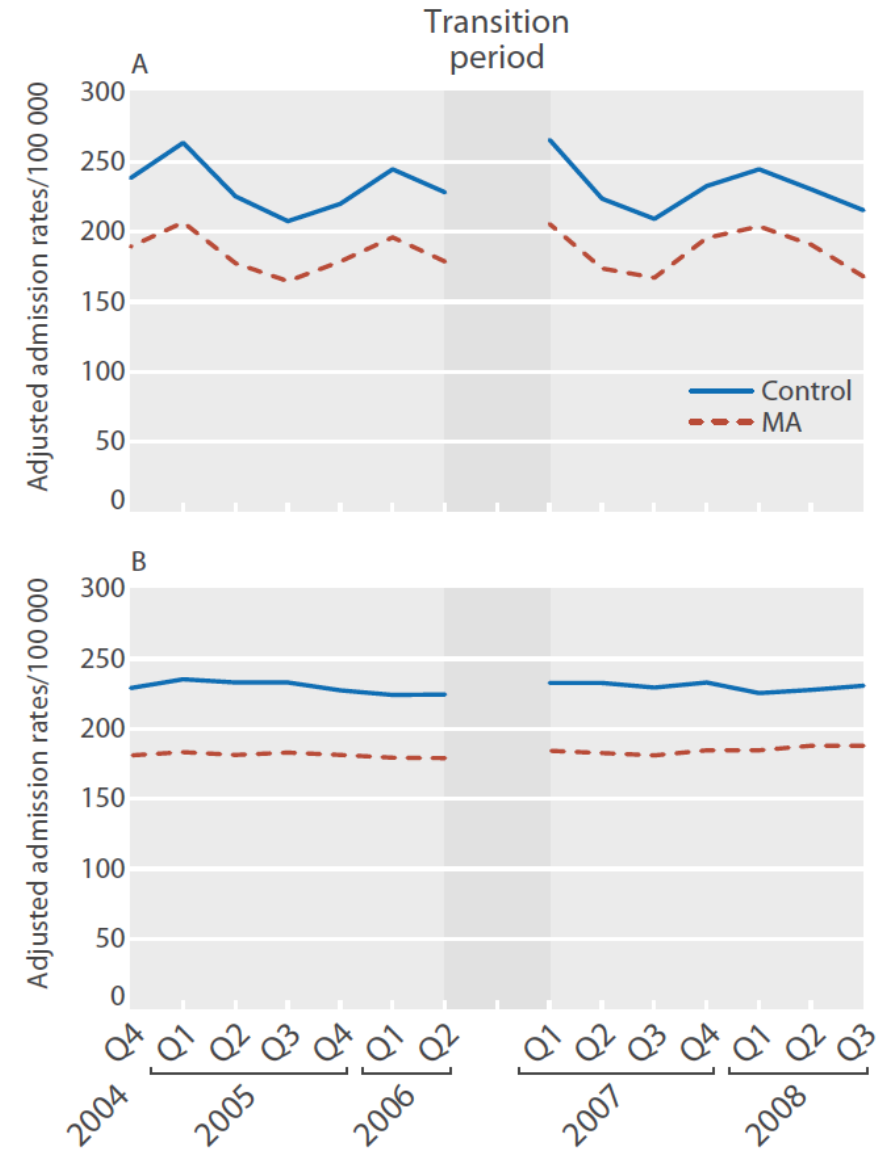
## Effect of Massachusetts healthcare reform on racial and ethnic disparities in admissions to hospital for ambulatory care sensitive conditions: retrospective analysis of hospital episode statistics

Danny McCormick,<sup>1</sup> Amresh D Hanchate,<sup>2, 3</sup> Karen E Lasser,<sup>3</sup> Meredith G Manze,<sup>3</sup> Mengyun Lin,<sup>3</sup> Chieh Chu,<sup>3</sup> Nancy R Kressin<sup>2, 3</sup>

- Evaluated impact of MA reform on inequalities in hospital admissions.
- Compared MA to nearby states: NY, NJ, PA.
- Intervention “worked”: % uninsured halved (12% to 6%) from 2004-06 to 2008-09.

## Evaluating pre-intervention trends

- Adds credibility to assumption that post-intervention trends **would have been similar** in the absence of the intervention.
- “Null” results help focus on alternative mechanisms linking disadvantage to hospital admissions.



# Synthetic Controls

# Synthetic control methods

- Inference from comparative case studies is limited if we cannot identify a control to represent the counterfactual scenario.
- Abadie and Gardeazabal ([2003](#)) pioneered the synthetic control method to examine the economic impact of terrorism in the Basque country, using other Spanish regions as control groups.
- The synthetic control method uses a data driven approach to compare the trend of an outcome in a treated unit with the trend in a synthetic composite area (the “synthetic control”).

# What is a synthetic control?

- A synthetic control is a weighted average of available control units that approximates the most relevant characteristics of the treated unit prior to the treatment
- The synthetic control mimics the values of the predictors of the outcome, including pre-intervention values of the outcome, for the treated unit before the intervention occurred
- The synthetic control represents the counterfactual scenario for a treated unit in the absence of the intervention under scrutiny
- Intuition: A weighted combination of comparison units (the “synthetic control”) provides a better comparison for the treated unit than any single comparison unit alone

ORIGINAL ARTICLE

Open Access

# Does a ban on trans fats improve public health: synthetic control evidence from Denmark



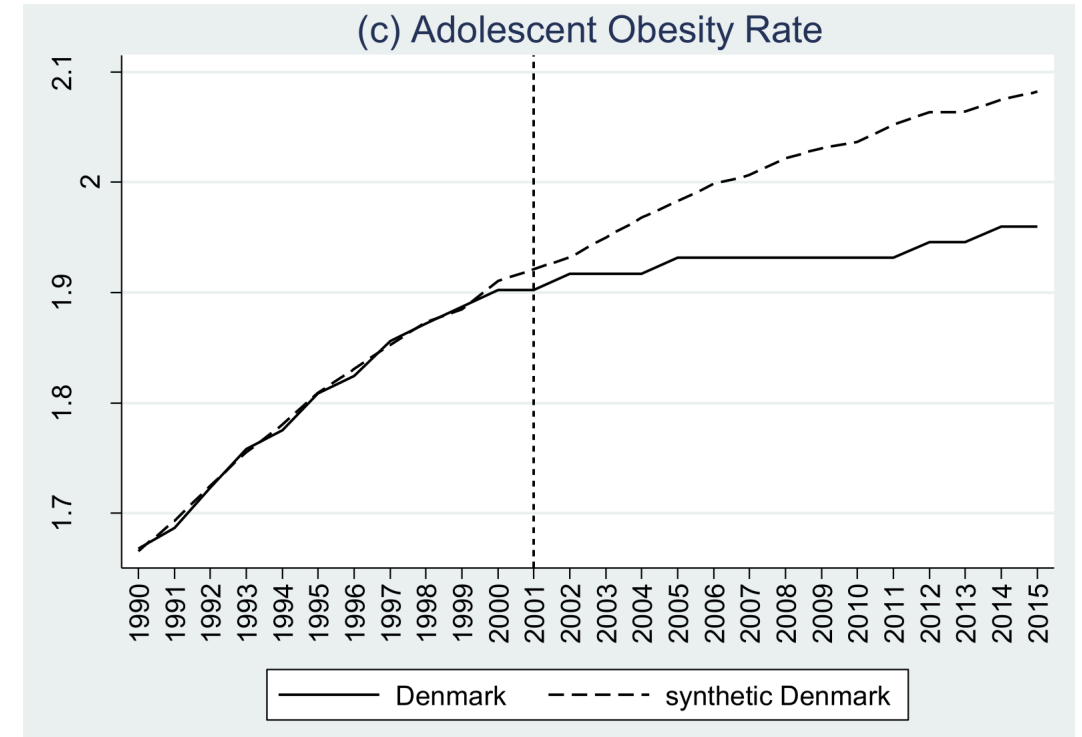
Rok Spruk\*  and Mitja Kovac

## Abstract

We examine the impact of the trans fat ban on a variety of public health outcomes. To this end, we consider a de facto trans fat ban that was introduced by Denmark in 2001. Using the synthetic control method, parallel trends between Denmark and countries in a control group in the years prior to the ban are used to construct a “synthetic Denmark” without any such trans fat ban. Our synthetic control estimates suggest the ban led to substantial improvements in public health. Following the ban, cardiovascular mortality dropped considerably, while the trends of adolescent and child obesity came to a halt and decreased significantly compared to the synthetic control group. Our findings provide new insights into the benefits for public health arising from the banning of trans fats.

- ‘Synthetic’ DK mostly SWE, ITA, USA, and FIN.
- Also declines in CVD mortality.
- Robustness checks:

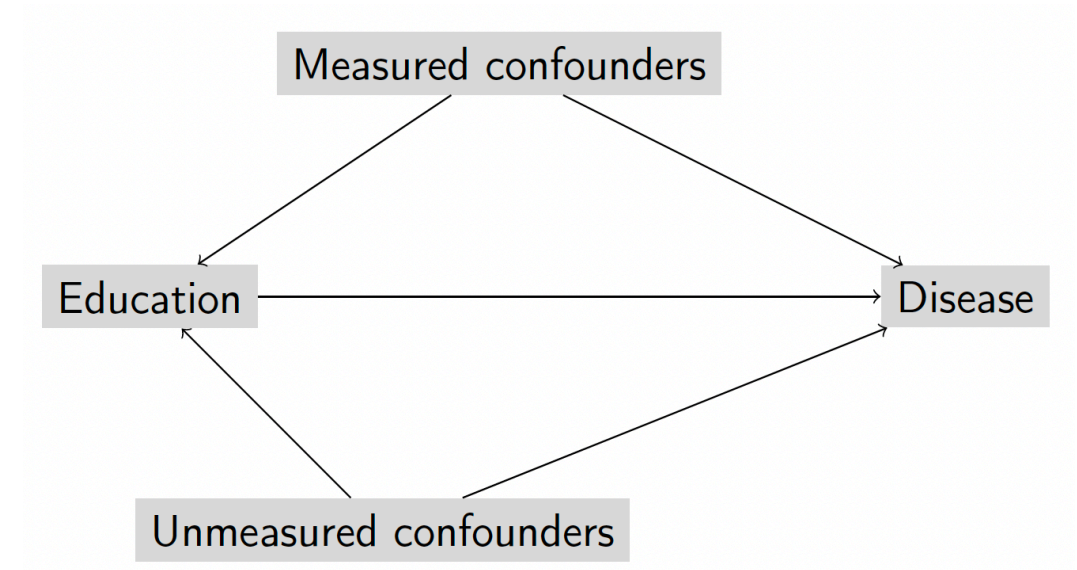
By deliberately assigning TFA policy to wrong dates and other countries, we show the effect of the 2001 TFA policy intervention is specific to Denmark and does not appear to be driven by alternative dates



# Instrumental Variables

# Challenge of conventional observational study (again)

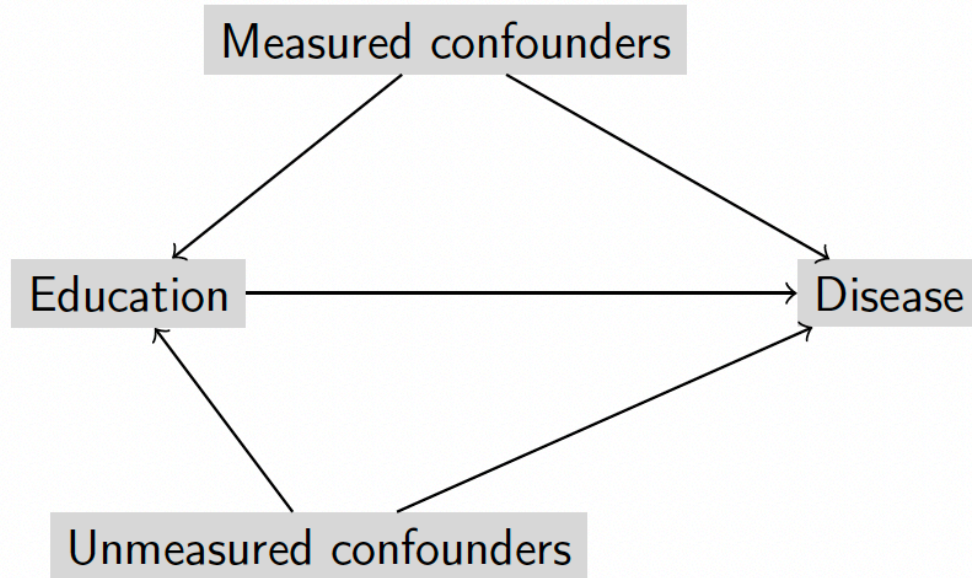
- Everyone: “Educational attainment is linked to improved health outcomes.”
- But what about unmeasured confounding? Unmeasured factors such as personality traits, cognitive ability, etc. may be predictive of both education and disease.
- Failure to measure such factors will falsely attribute their effects to education.



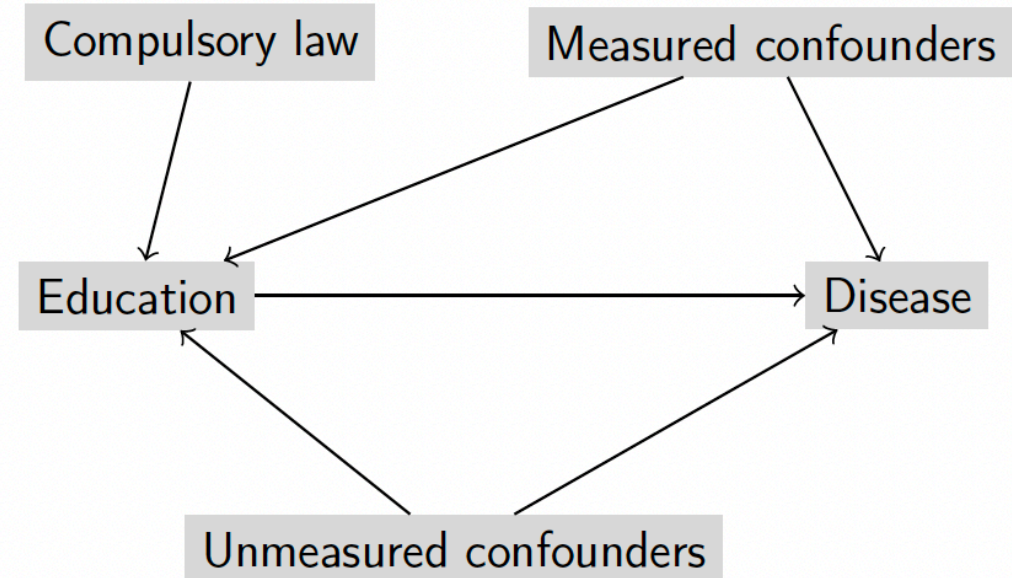
## Possible solution: Quasi-experiment

“Instrumental variable”: predicts education but **not** associated with anything else [mimicking random assignment].

Observational study

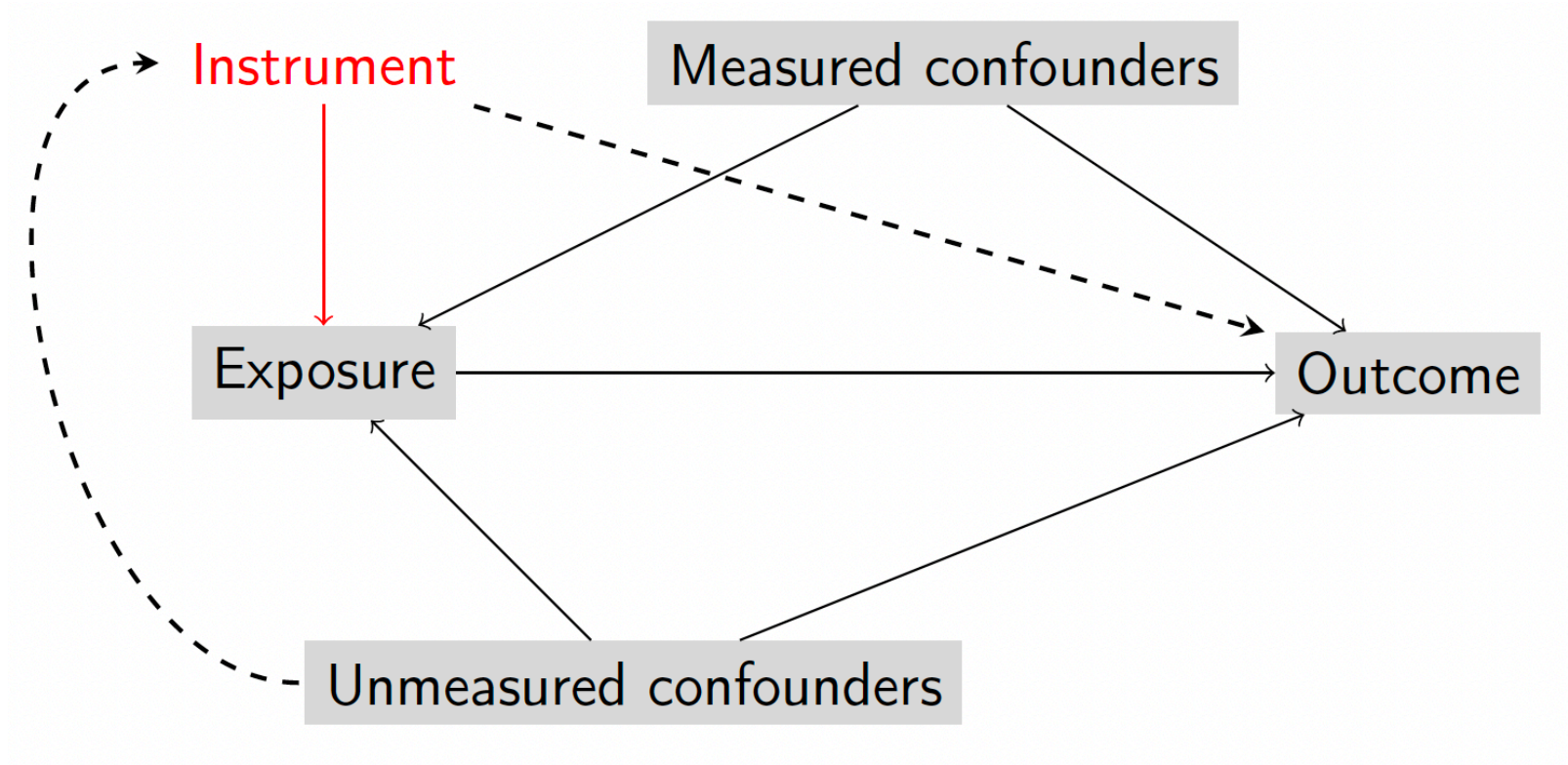


Quasi-experimental study



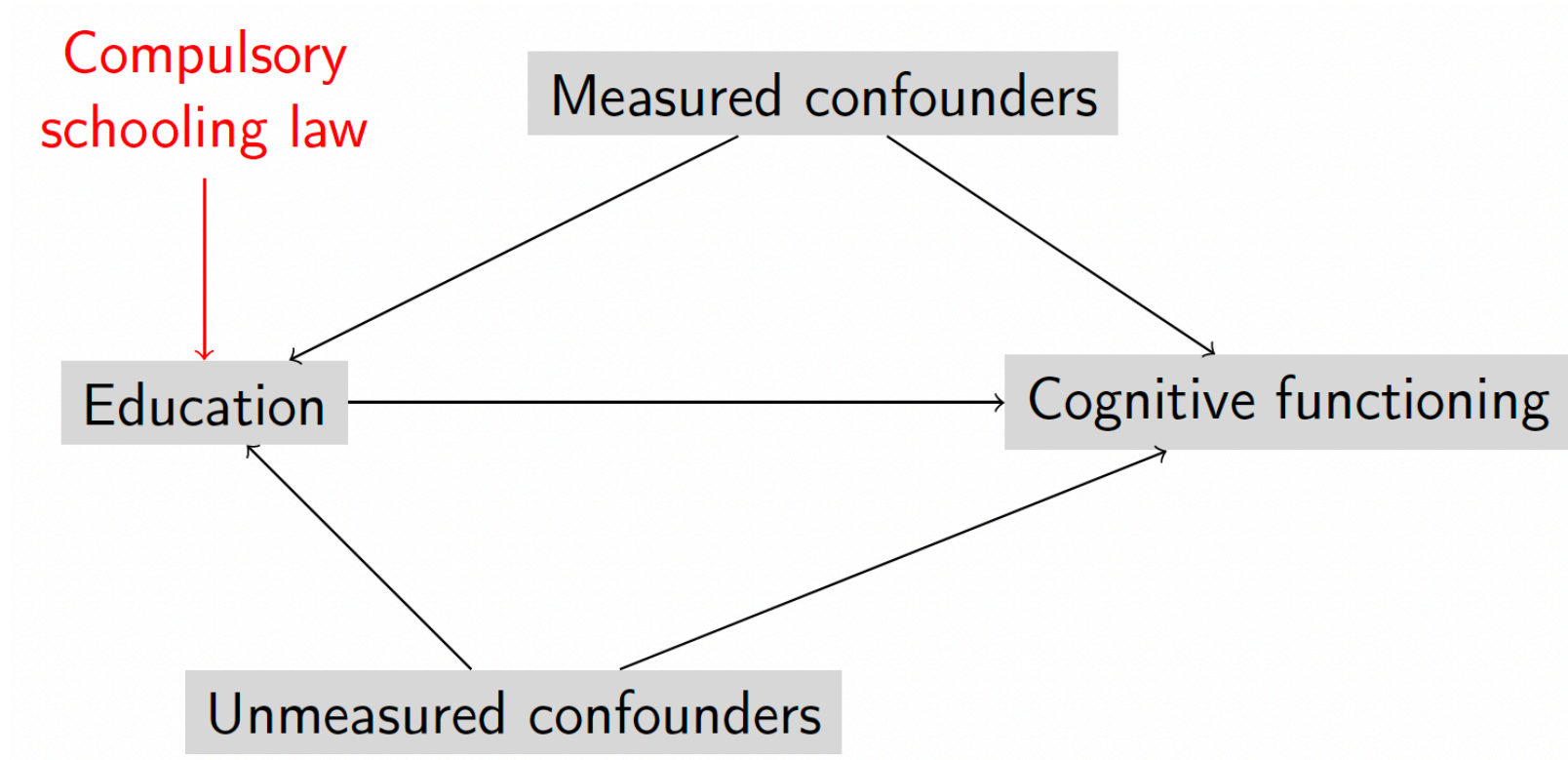
## Non-randomized instrument creates additional issues

- In an RCT we know the treatment assignment is not associated directly with the outcome or with other unmeasured common causes.
- This assumption is less credible when our “instrument” is non-randomized.



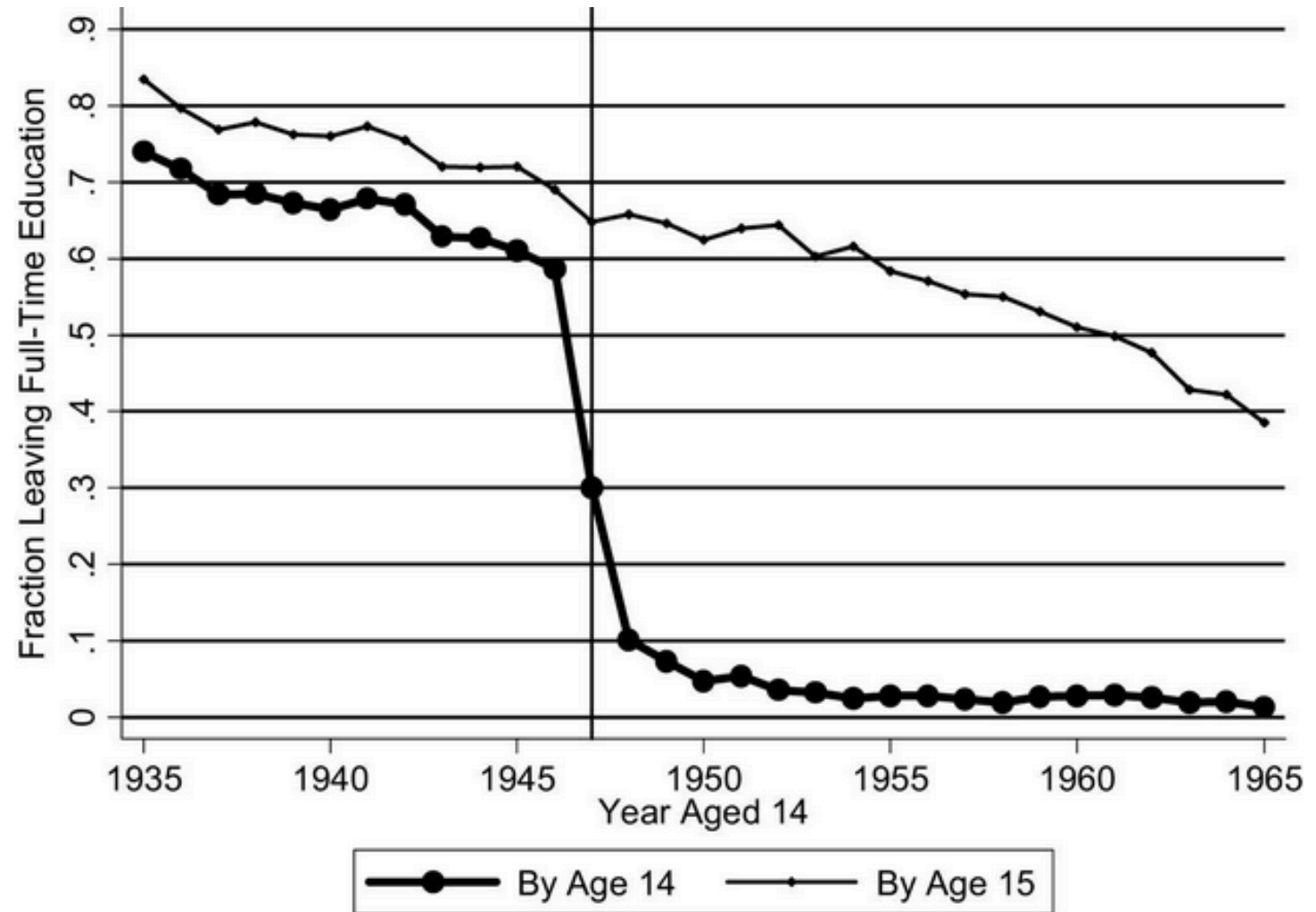
## Non-randomized examples of IV: Policies

- Does education affect cognitive functioning?
- Instrument: changes in compulsory schooling laws [mimicking random assignment].



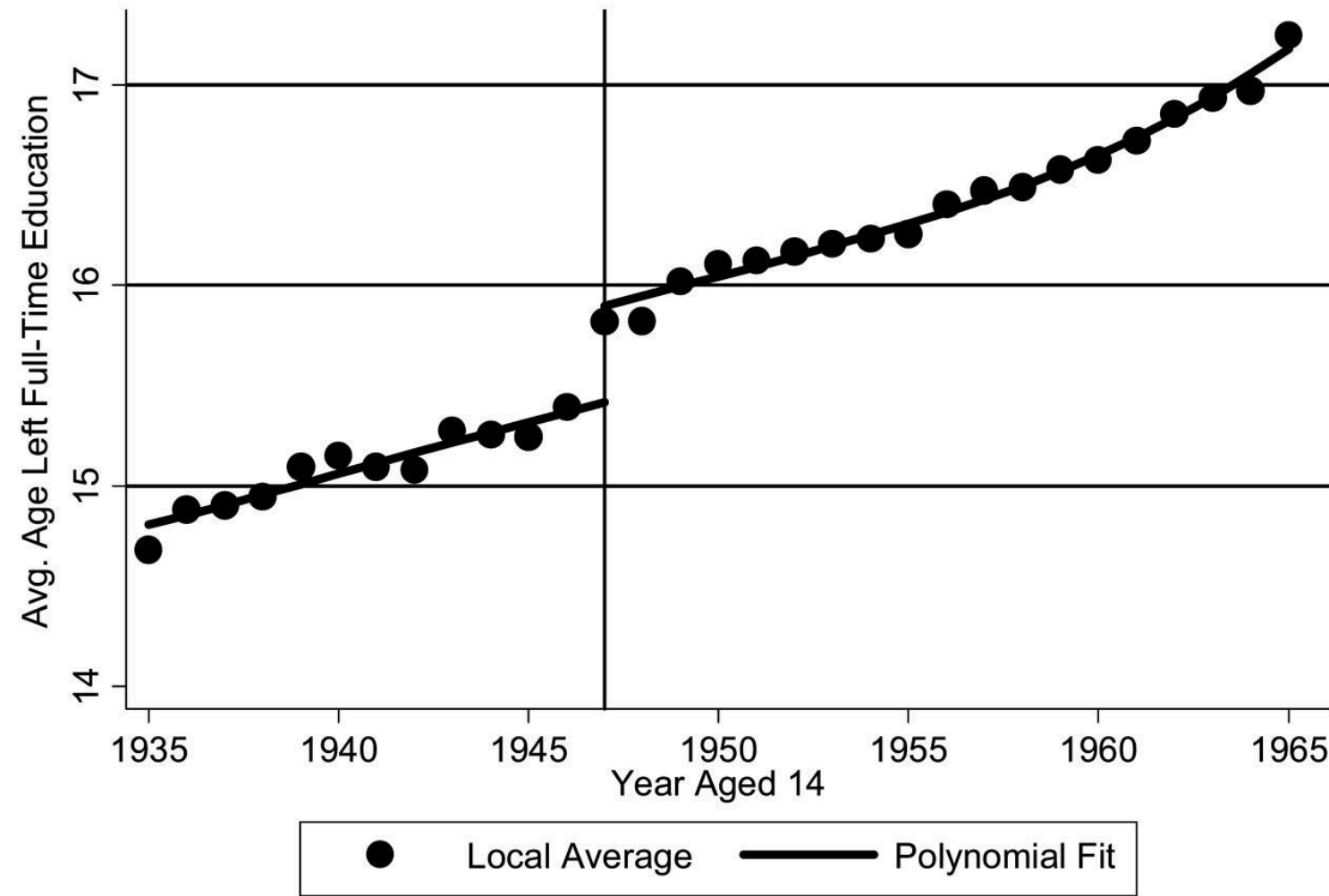
What does a quasi-experiment look like?

Fraction left full-time education by year aged 14 and 15 (Great Britain)



The lower line shows the proportion of British-born adults aged 32 to 64 from the 1983 to 1998 General Household Surveys who report leaving full-time education at or before age 14 from 1935 to 1965. The upper line shows the same, but for age 15. The minimum school-leaving age in Great Britain changed in 1947 from 14 to 15 ([Oreopoulos 2006](#)).

Average schooling increases by exactly half a year between the cohorts that were age 14 in 1946 and in 1948.



# Length of secondary schooling and risk of HIV infection in Botswana: evidence from a natural experiment

Jan-Walter De Neve, Günther Fink, S V Subramanian, Sikhulile Moyo, Jacob Bor



## Summary

**Background** An estimated 2·1 million individuals are newly infected with HIV every year. Cross-sectional and longitudinal studies have reported conflicting evidence for the association between education and HIV risk, and no randomised trial has identified a causal effect for education on HIV incidence. **We aimed to use a policy reform in secondary schooling in Botswana to identify the causal effect of length of schooling on new HIV infection.**

**Methods** Data for HIV biomarkers and demographics were obtained from the nationally representative household 2004 and 2008 Botswana AIDS Impact Surveys (N=7018). In 1996, Botswana reformed the grade structure of secondary school, expanding access to grade ten and increasing educational attainment for affected cohorts. Using exposure to the policy reform as an instrumental variable, we used two-stage least squares to estimate the causal effect of years of schooling on the cumulative probability that an individual contracted HIV up to their age at the time of the survey. We also assessed the cost-effectiveness of secondary schooling as an HIV prevention intervention in comparison to other established interventions.

**Findings** **Each additional year of secondary schooling caused by the policy change led to an absolute reduction in the cumulative risk of HIV infection of 8·1 percentage points ( $p=0\cdot008$ ),** relative to a baseline prevalence of 25·5% in the pre-reform 1980 birth cohort. Effects were particularly large in women (11·6 percentage points,  $p=0\cdot046$ ). Results were robust to a wide array of sensitivity analyses. Secondary school was cost effective as an HIV prevention intervention by standard metrics (cost per HIV infection averted was US\$27 753).

**Interpretation** Additional years of secondary schooling had a large protective effect against HIV risk in Botswana, particularly for women. Increasing progression through secondary school could be a cost-effective HIV prevention measure in HIV-endemic settings, in addition to yielding other societal benefits.

**Funding** Takemi Program in International Health at the Harvard T.H.Chan School of Public Health, Belgian American Educational Foundation, Fernand Lazard Foundation, Boston University, National Institutes of Health.

*Lancet Glob Health* 2015;  
3: e470–77

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See [Comment](#) page e428

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Correspondence to: Jacob Bor, Department of Global Health, Boston University School

# Regression Discontinuity

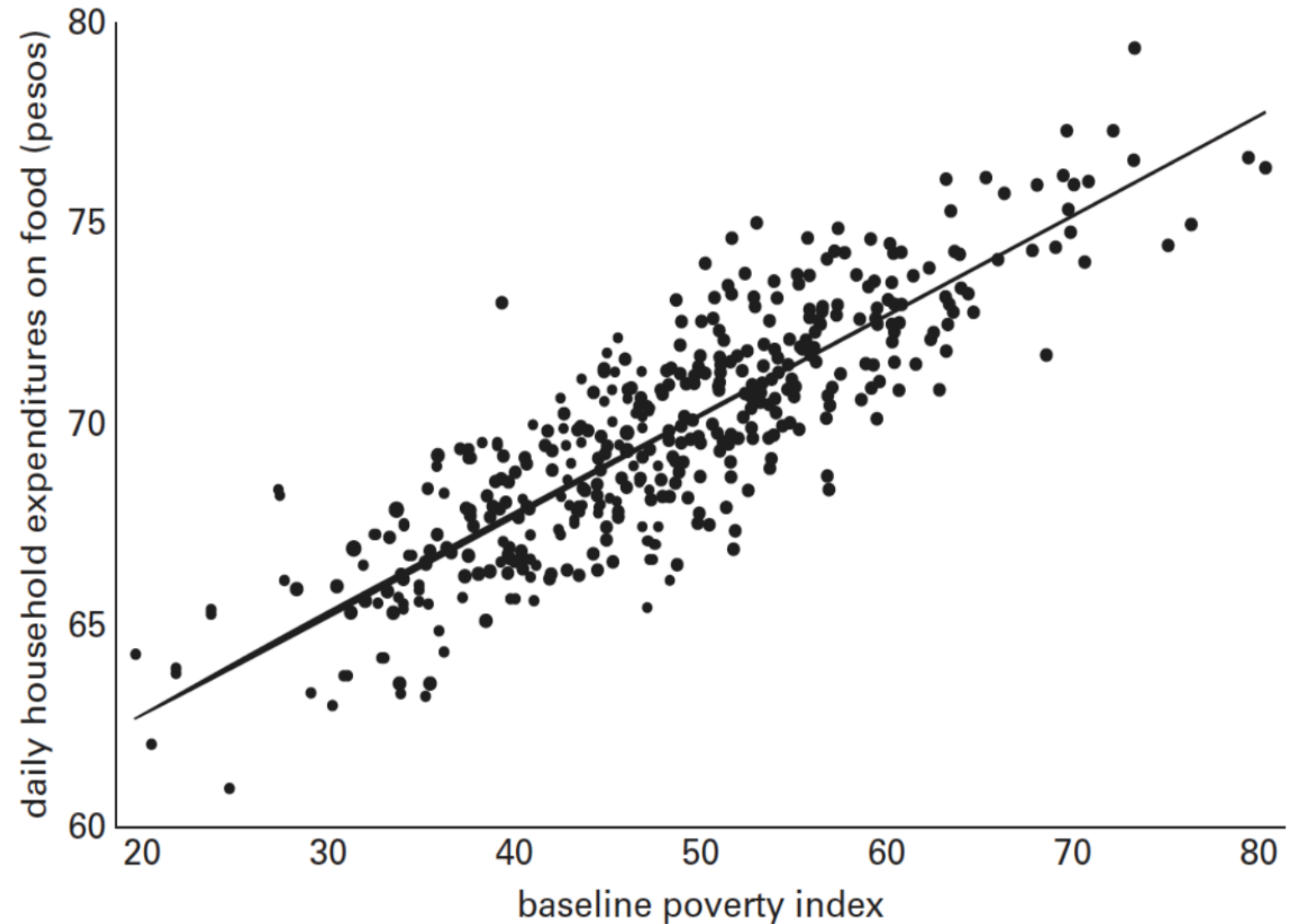
## RD: Basic Idea

- Take advantage of arbitrary thresholds that sometimes assign treatment to individuals.
- When an administrative or rule-based cutoff in a continuous variable (present in your data) predicts treatment assignment, being on one side or the other of this cutoff determines, or predicts, treatment received.
- The continuous variable is called the “assignment” or “forcing” variable.
- Groups just on either side are the threshold considered “as good as randomly” assigned to treatment.

## RD: Motivating example

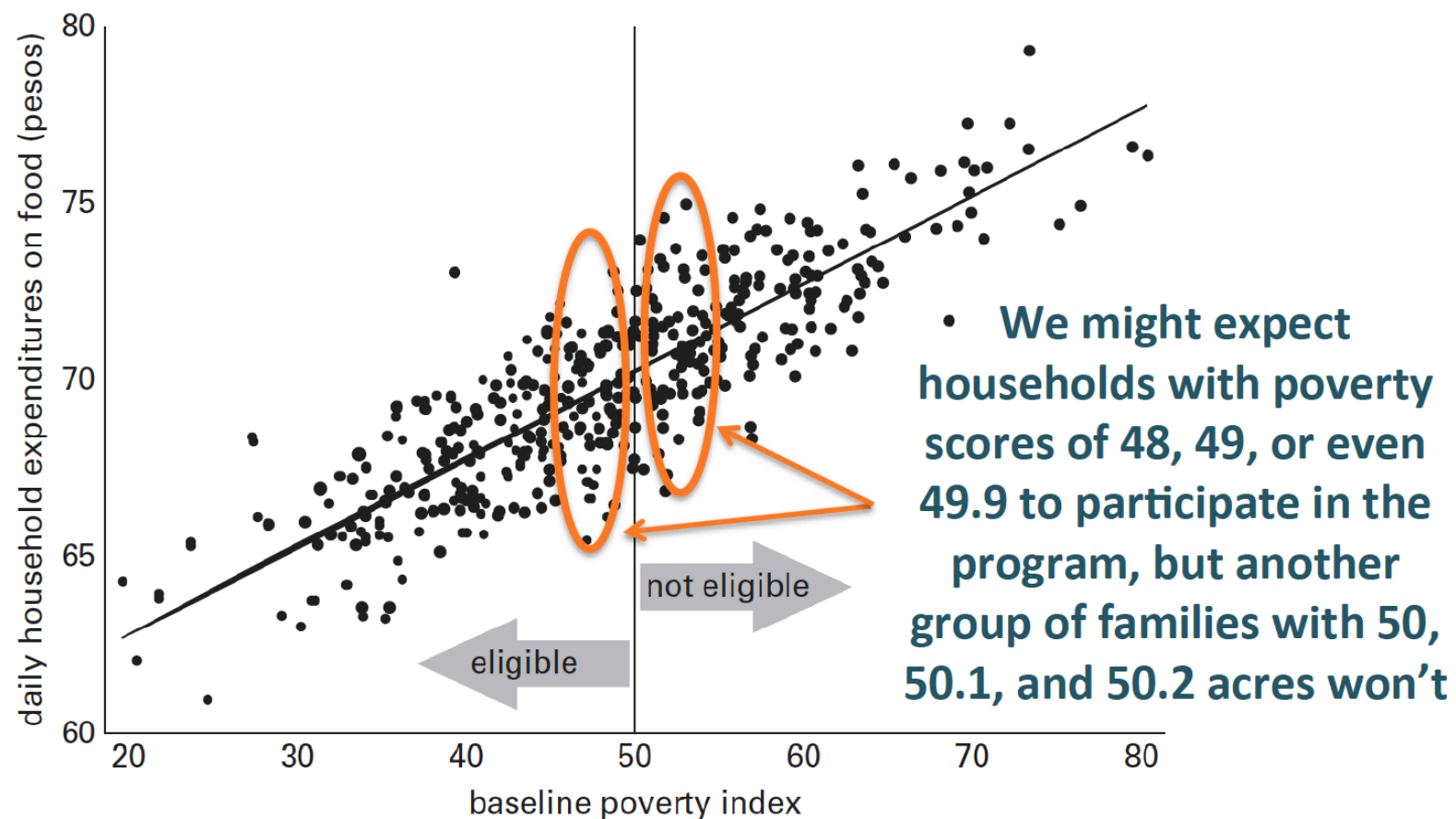
- Suppose we want to estimate the impact of a cash transfer program on daily food expenditure of poor households.
- Poverty is measured by a continuous score between 0 and 100 that is used to rank households from poorest to richest.
- Poverty is the assignment variable,  $Z$ , that determines eligibility for the cash transfer program.
- The outcome of interest, daily food expenditure, is denoted by  $Y$ .

At baseline, you might expect poorer households to spend less on food, on average, than richer ones, which might look like this

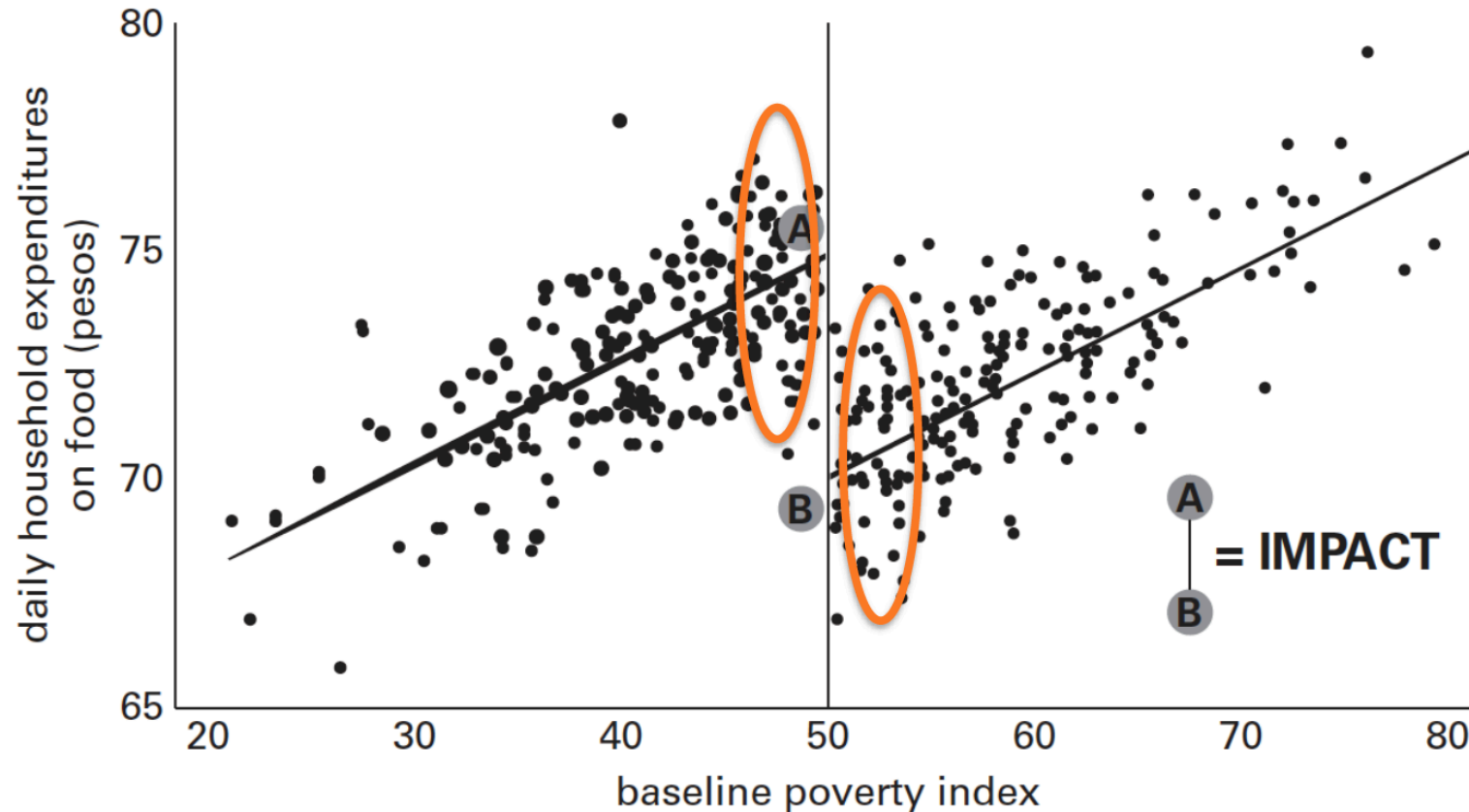


Gertler et al. ([2011](#))

Under the program's rules, only households with a poverty score,  $Z$ , below 50 are eligible for the cash payment



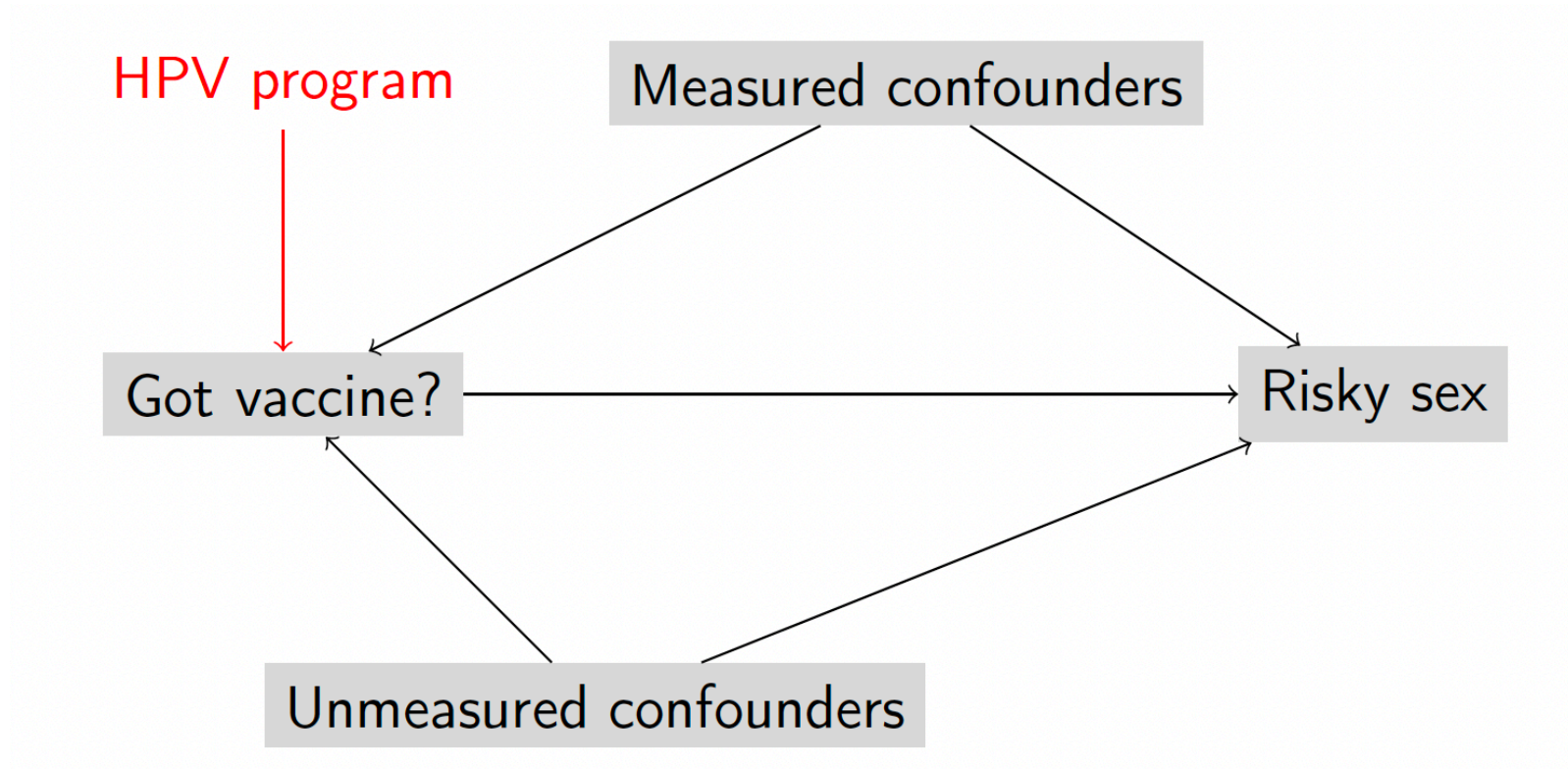
As we approach the cutoff value from above and below, the individuals in both groups become more and more alike, on both measured and unobserved characteristics — in a small area around the threshold, the only difference is in treatment assignment



Gertler et al. (2011)

# Applied example: HPV vaccine and sexual behaviors

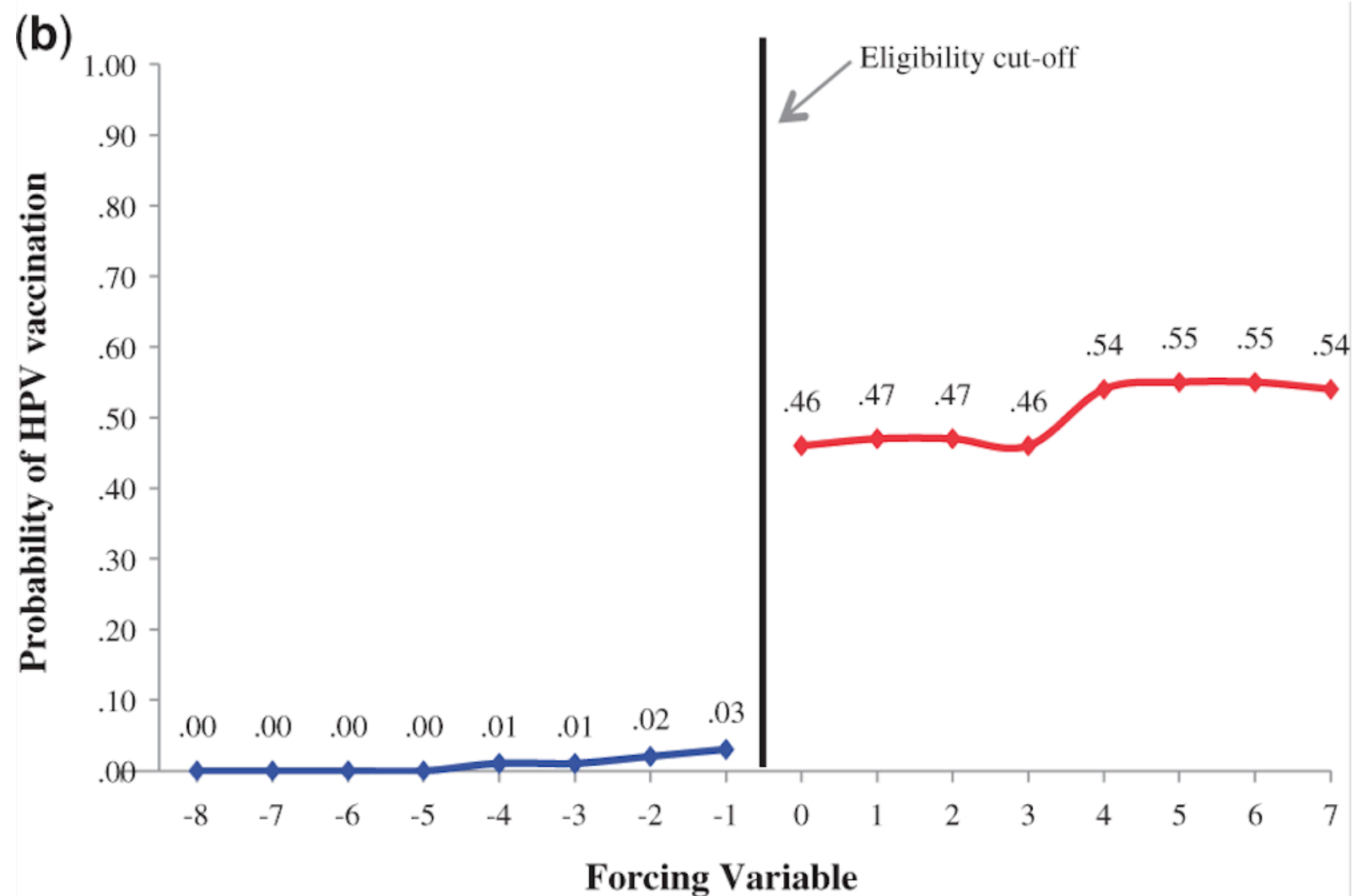
- Does getting the HPV vaccine affect sexual behaviors?
- Vaccine policy: predicts vaccine receipt but (we assume) **not** associated with anything else [mimicking random assignment].



Smith et al. ([2015](#))

Does the cutoff predict treatment?

- Girls “assigned” to HPV program by quarter of birth.
- $\Pr(\text{vaccine})$  jumps discontinuously at cutoff



Smith et al. (2015)

What does a credible natural experiment look like?

| <b>Table 1:</b> Baseline characteristics of the eligibility groups in the study cohort |                                                       |                           |                                 |                                                       |                           |
|----------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------|---------------------------------|-------------------------------------------------------|---------------------------|
| Characteristic                                                                         | Program eligibility group;<br>% of eligibility group* |                           | Characteristic                  | Program eligibility group;<br>% of eligibility group* |                           |
|                                                                                        | Ineligible<br>(n = 131 781)                           | Eligible<br>(n = 128 712) |                                 | Ineligible<br>(n = 131 781)                           | Eligible<br>(n = 128 712) |
| <b>Sociodemographic†</b>                                                               |                                                       |                           | <b>Health services use***††</b> |                                                       |                           |
| Age, yr, mean ± SD                                                                     | 13.17 ± 0.28                                          | 13.17 ± 0.28              | Hospital admission              |                                                       |                           |
| Birth quarter                                                                          |                                                       |                           | 0                               | 98.0                                                  | 98.2                      |
| Jan.–Mar.                                                                              | 24.3                                                  | 24.2                      | ≥ 1                             | 2.0                                                   | 1.8                       |
| Apr.–June                                                                              | 26.1                                                  | 26.1                      | LOS, d, mean ± SD               | 7.4 ± 15.6                                            | 8.0 ± 18.2                |
| July–Sept.                                                                             | 25.7                                                  | 25.8                      | Same-day surgery                |                                                       |                           |
| Oct.–Dec.                                                                              | 23.9                                                  | 23.9                      | 0                               | 97.7                                                  | 97.8                      |
| Residency                                                                              |                                                       |                           | ≥ 1                             | 2.4                                                   | 2.2                       |
| Urban                                                                                  | 85.3                                                  | 85.8                      | Emergency department visits     |                                                       |                           |
| Rural                                                                                  | 14.0                                                  | 13.5                      | 0                               | 70.7                                                  | 71.1                      |
| Missing‡                                                                               | 0.7                                                   | 0.6                       | 1                               | 18.1                                                  | 17.8                      |
| Income quintile                                                                        |                                                       |                           | ≥ 2                             | 11.2                                                  | 11.1                      |
| 1 (lowest)                                                                             | 16.6                                                  | 15.0                      | Outpatient visits               |                                                       |                           |
| 2                                                                                      | 18.4                                                  | 17.8                      | 0 or 1                          | 22.6                                                  | 22.8                      |
| 3                                                                                      | 20.6                                                  | 21.1                      | 2–5                             | 27.4                                                  | 26.9                      |
| 4                                                                                      | 22.0                                                  | 23.1                      | 6–12                            | 25.1                                                  | 24.5                      |
| 5 (highest)                                                                            | 21.4                                                  | 22.1                      | ≥ 13                            | 25.0                                                  | 25.8                      |

Smith et al. (2015)

Note little impact of adjustment

| <b>Table 3:</b> Effect of quadrivalent human papillomavirus vaccination on clinical indicators of sexual behaviour*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                             |                     |                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------|-----------------------|
| Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | No. of excess cases per 1000 girls (95% CI) | RR (95% CI)         | Adjusted† RR (95% CI) |
| <b>Effect of vaccine</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                             |                     |                       |
| Composite outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | −0.61 (−10.71 to 9.49)                      | 0.96 (0.81 to 1.14) | 0.98 (0.84 to 1.14)   |
| Pregnancy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.70 (−7.57 to 8.97)                        | 0.99 (0.79 to 1.23) | 1.00 (0.83 to 1.21)   |
| STIs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | −4.92 (−11.49 to 1.65)                      | 0.81 (0.62 to 1.05) | 0.81 (0.63 to 1.04)   |
| <b>Effect of program</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                             |                     |                       |
| Composite outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | −0.25 (−4.35 to 3.85)                       | 0.99 (0.93 to 1.06) | 1.00 (0.93 to 1.07)   |
| Pregnancy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.29 (−3.07 to 3.64)                        | 1.00 (0.92 to 1.09) | 1.01 (0.93 to 1.10)   |
| STIs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | −2.00 (−4.67 to 0.67)                       | 0.92 (0.83 to 1.03) | 0.92 (0.83 to 1.03)   |
| <p>Note: CI = confidence interval, RR = relative risk, STIs = sexually transmitted infections.</p> <p>*To address the effect of birth timing that we observed, we used the entire bandwidth of data (i.e., all observations in the 1992 to 1995 birth cohorts) and included birth quarter as a covariate in the model. In all analyses, the birth cohorts closest to the cut-off (1993 and 1994) were weighted twice as heavily as those furthest from the cut-off (1992 and 1995).</p> <p>†In this sensitivity analysis, we adjusted for neighbourhood income quintile, hepatitis B vaccination and history of sexual health-related indicator, as well as for birth quarter.</p> |                                             |                     |                       |

Smith et al. (2015)

## Issues related to generalizability

- RD estimates local average impacts around the eligibility cutoff where treated and control units are most similar and results cannot be generalized to units whose scores are further away from the cutoff (unless we assume treatment heterogeneity).
- If the goal is to answer whether the program should exist or not, then RD is likely not the appropriate methodology.
- However, if the question is whether the program should be cut or expanded at the margin, then it produces the local estimate of interest to inform this policy decision

# Overview

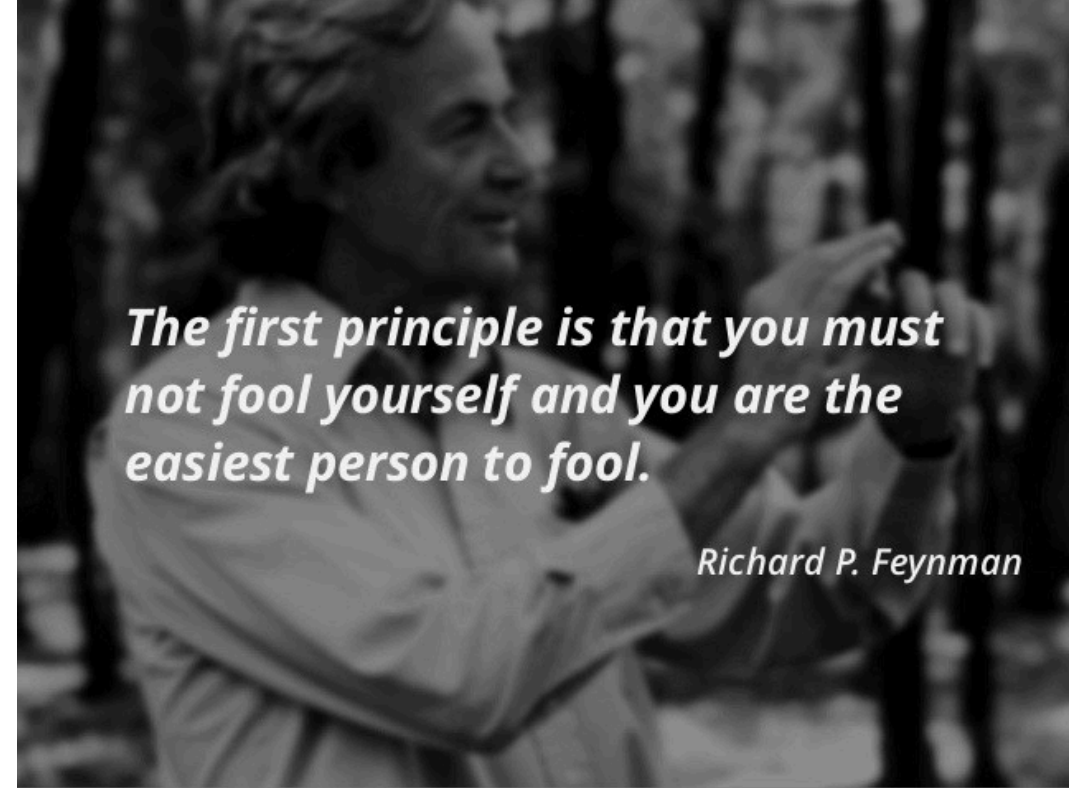
Motivation

Quasi-Experimental Designs

Final Thoughts

## Be careful, and skeptical

- Correlations between social factors and health are easy to find.
- They do not necessarily reflect causal relationships.
- Need to search hard for alternative explanations.
- Important to consider the strength of evidence in considering interventions.

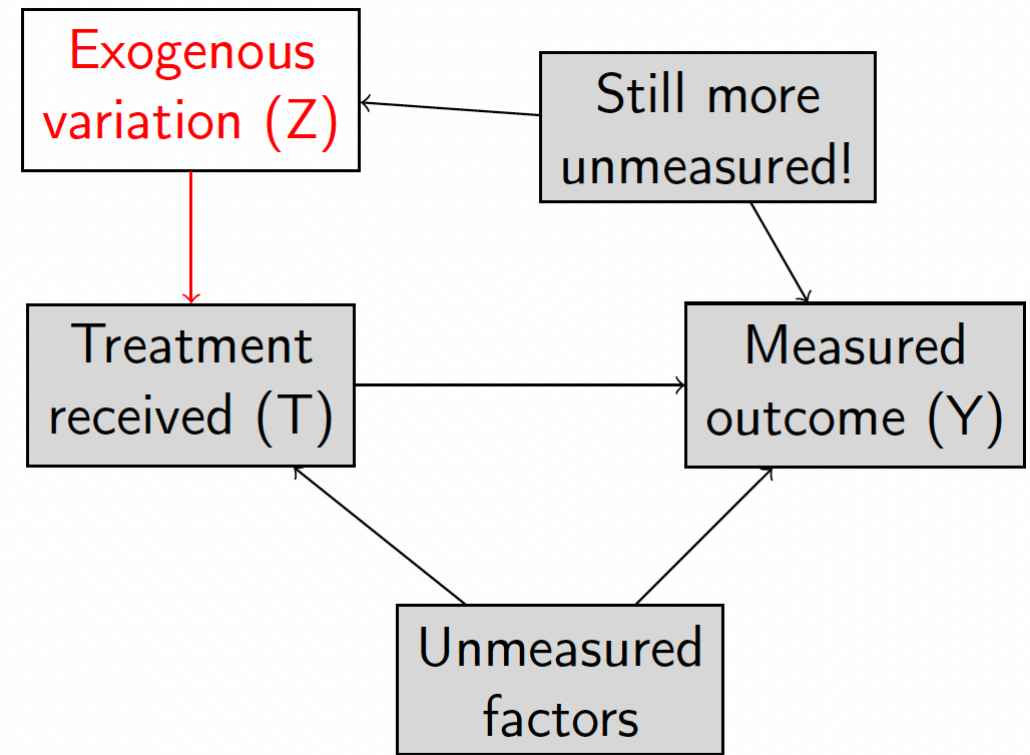


# Are natural experiments always more credible?

- Not necessarily, but probably.
- Key is “as-if” randomization of treatment:
  - If this is credible, it is a much stronger design than most observational studies.
  - Should eliminate self-selection in to exposure groups.
- Allows for simple, transparent analysis of average differences between groups.
- Allows us to rely on weaker assumptions.

# Assumptions still matter!

- Quasi-experimental studies are still observational.
- Most credible if they create unconditional randomized treatment groups (e.g., lottery).
- Credibility is continuous, not binary.



## Back to basics: assumptions and costs

- Major benefit of randomized evaluations are that few assumptions are needed to estimate a causal effect.
- Necessary assumptions can often be checked.
- Non-randomization means more assumptions, more possibility for assumptions to be violated.
- Should lead us to spend lots of time trying to test the credibility of these assumptions.
  - How good is “as-if random”?
  - Are there compelling non-causal alternative explanations for the observed results?
- Not all non-randomized designs are created equal.

# References

- Abadie, Alberto, and Javier Gardeazabal. 2003. "The Economic Costs of Conflict: A Case Study of the Basque Country." *American Economic Review* 93 (1): 113–32. <https://doi.org/10.1257/000282803321455188>.
- Carroll, Aaron E. 2018. "Workplace Wellness Programs Don't Work Well. Why Some Studies Show Otherwise." *The New York Times*.
- De Neve, Jan-Walter, Günther Fink, S. V. Subramanian, Sikhulile Moyo, and Jacob Bor. 2015. "Length of Secondary Schooling and Risk of HIV Infection in Botswana: Evidence from a Natural Experiment." *The Lancet Global Health* 3 (8): e470–77. [https://doi.org/10.1016/S2214-109X\(15\)00087-X](https://doi.org/10.1016/S2214-109X(15)00087-X).
- Dunning, Thad. 2012. *Natural Experiments in the Social Sciences: A Design-Based Approach*. Cambridge: Cambridge University Press.
- Gertler, Paul J., Sebastian Martinez, Patrick Premand, Laura B. Rawlings, and Christel M. J. Vermeersch. 2011. *Impact Evaluation in Practice*. Washington, DC: The World Bank.
- Glymour, M. Maria, Ichiro Kawachi, Christopher S. Jencks, and Lisa F. Berkman. 2008. "Does Childhood Schooling Affect Old Age Memory or Mental Status? Using State Schooling Laws as Natural Experiments." *Journal of Epidemiology & Community Health* 62 (6): 532–37. <https://doi.org/10.1136/jech.2006.059469>.
- Jones, Damon, David Molitor, and Julian Reif. 2019. "What Do Workplace Wellness Programs Do? Evidence from the Illinois Workplace Wellness Study." *The Quarterly Journal of Economics* 134 (4): 1747–91. <https://doi.org/10.1093/qje/qjz023>.
- Keall, Michael D., Nevil Pierse, Philippa Howden-Chapman, Chris Cunningham, Malcolm Cunningham, Jagadish Guria, and Michael G. Baker. 2015. "Home Modifications to Reduce Injuries from Falls in the Home Injury Prevention Intervention (HIPI) Study: A Cluster-Randomised Controlled Trial." *The Lancet* 385 (9964): 231–38. [https://doi.org/10.1016/S0140-6736\(14\)61006-0](https://doi.org/10.1016/S0140-6736(14)61006-0).
- McCormick, Danny, Amresh D. Hanchate, Karen E. Lasser, Meredith M. Manze, Mengyun Lin, Chieh Chu, and Nancy R. Kressin. 2015. "Effect of Massachusetts Healthcare Reform on Racial and Ethnic Disparities in Admissions to Hospital for Ambulatory Care Sensitive Conditions: Retrospective Analysis of Hospital Episode Statistics." *BMJ* 350: h1480. <https://doi.org/10.1136/bmj.h1480>.
- Oreopoulos, Philip. 2006. "Estimating Average and Local Average Treatment Effects of Education When Compulsory Schooling Laws Really Matter." *American Economic Review* 96 (1): 152–75. <https://doi.org/10.1257/000282806776157641>.
- Smith, Leah M., Jay S. Kaufman, Erin C. Strumpf, and Linda E. Lévesque. 2015. "Effect of Human Papillomavirus (HPV) Vaccination on Clinical Indicators of Sexual Behaviour Among Adolescent Girls: The Ontario Grade 8 HPV Vaccine Cohort Study." *Canadian Medical Association Journal* 187 (2): E74–81. <https://doi.org/10.1503/cmaj.140900>.
- Spruk, Rok, and Mitja Kovač. 2020. "Does a Ban on Trans Fats Improve Public Health: Synthetic Control Evidence from Denmark." *Swiss Journal of Economics and Statistics* 156 (1): 4. <https://doi.org/10.1186/s41937-019-0048-0>.